

- (iv) Calculate the electric field strength  $E$  at the surface of the sphere. Give a unit with your answer.

$E =$  ..... unit ..... [2]

- (c) The sphere in (b) is discharged by connecting it to earth (0 V) through a resistor of resistance  $120 \text{ M}\Omega$ .

Calculate the time taken for the charge to fall to  $26 \text{ pC}$ .

time = ..... s [2]

[Total: 11]

